



A glimpse of Planet Labs' satellites

small satellites of all scales, private companies are also investing heavily in this area and are now expecting great returns by contracting earth imagery data to customers.

Talking about future plans at Planet Labs, Schingler reports, "We are launching a commercial constellation of 28 satellites at the end of the year. They will be taking photos of the earth. We hope to accurately assess changing landscapes. Our imagery will be an entirely new data layer. We hope to innovate at every part of the product to streamline delivery of imagery to our partners and customers. Currently, the average person doesn't know what he or she can do with frequently-updated information about the changing planet. This shift in the industry will open up satellite imagery to more engineers and small tech firms."

After successfully launching two experimental satellites, Planet Labs is confident about launching the 28-satellite constellation together on an Antares rocket, a new launcher built by the US-based Orbital Sciences Corporation, by the end of this year. The CubeSat-based, solar-powered satellites will form a tilted ring around the earth, and orbit at an altitude of 400 km. Since the satellites have no propulsion systems, they are likely to lose speed in two to five years, after which they will fall towards Earth

and burn up, making way for a new constellation. While in operation, they will send their images to three or more ground stations, where the data will be processed and uploaded for customers to use.

The other key player in this space, is US-based Skybox Imaging, an information and analytics company that also promises reliable and frequent high-resolution imagery including what it claims to be the first-ever HD video of the earth. Skybox will be using a constellation of over 24 microsatellites, of which the first two are likely to be launched later this year using Virgin Galactic's LauncherOne launch vehicle. (According to Wikipedia, "Virgin Galactic is a company within Richard Branson's Virgin Group, which plans to provide suborbital spaceflights to space tourists, suborbital launches for space science missions and orbital launches of small satellites.")

With small satellites becoming so popular, launch facilities are also emerging as a good business! Virgin Galactic's LauncherOne, which can launch payloads of 100 kg into low-earth orbit, is one of the most talked-about facilities. While it might take some months to get into operation, several US-based companies, including GeoOptics, Skybox Imaging, Spaceflight Services and Planetary

Resources, are said to have already contracted LauncherOne. Plus, Surrey Satellite Technology and Sierra Nevada Space Systems, the two largest manufacturers of small satellites, are also said to be developing satellite buses optimised to the design of LauncherOne. US-based Garvey Spacecraft, too, recently received funding to develop a clustered nano/microsatellite launch vehicle capable of delivering 20kg payloads into 450km circular orbits, based on its Prospector 18 suborbital launch vehicle technology. Boeing is also working on an air-launched three-stage-to-orbit launch vehicle concept capable of launching small payloads of 45 kg into low-earth orbit.

Students at the forefront

On the non-commercial front, a recent and revolutionary project that must be discussed is ArduSat, the first open source nanosatellite developed by Nanosatifi LLC, an aerospace company founded by four graduate students in the US in 2012. Based on the CubeSat standard, ArduSat contains a set of Arduino boards and sensors. ArduSat was successfully launched this August, with a stated goal of providing people open access to space! Priya Kuber, managing director of Arduino India, says, "ArduSat is an Arduino satellite wherein Arduino users can test their code in space. It is exciting to know that now Indian students would be able to have the same access to space! ArduSat has tied up with Dhruva Space India, and we are partnering with them to teach space-oriented experiments in a fun and educative way (<http://www.spaceschool.co.in>)."

"A few universities have developed small satellites with the help of ISRO. SRMSAT and YOUTH SAT, launched by ISRO's launch vehicles during 2011, are examples," John points out.

Anna University successfully launched the first student-built Indian satellite in 2009. Called Anusat, this is a 60×60×60cm cube-shaped